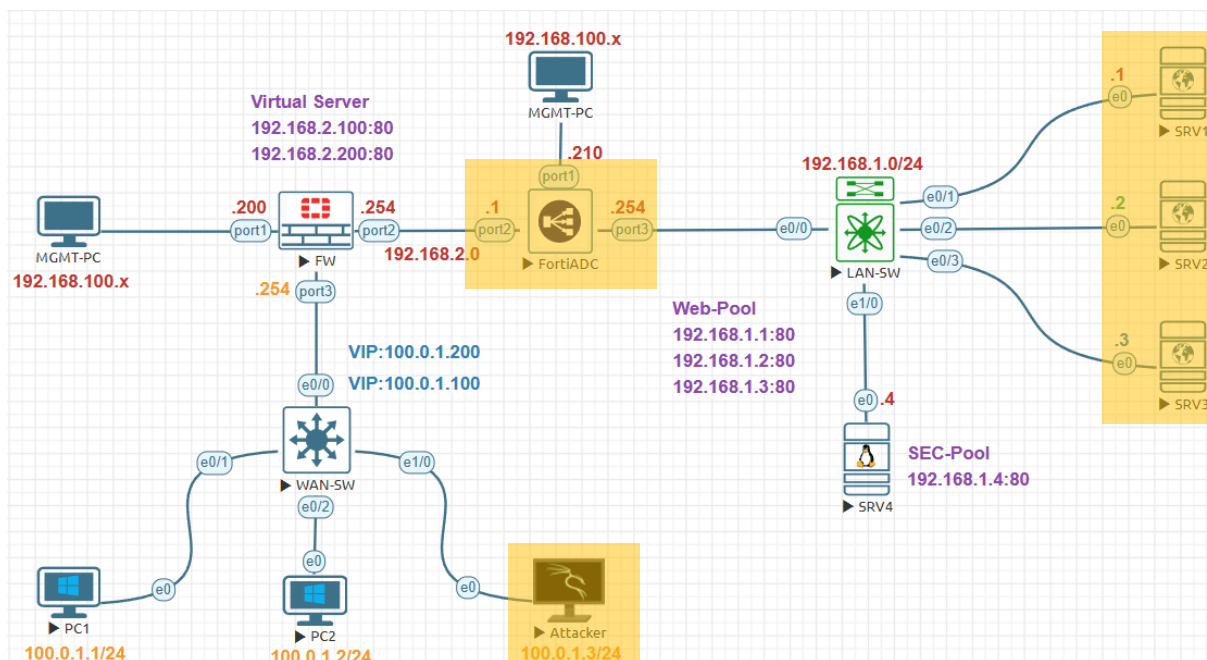


IP-Based Persistence Lab:



Management Subnet	192.168.100.0/24
FortiGate Management IP	192.168.100.200
FortiADC Management IP	192.168.100.210
Internal Servers Subnet	192.168.1.0/24
FortiADC External Subnet	192.168.2.0/24
FortiGate Firewall External	100.0.1.0/24
FortiADC Internal IP Address	192.168.1.254
FortiADC External IP Address	192.168.2.1
FortiGate Firewall Internal IP	192.168.2.254
FortiGate Firewall External IP	100.0.1.254
SRV1 IP Address	192.168.1.1
SRV2 IP Address	192.168.1.2
SRV3 IP Address	192.168.1.3
SRV3 IP Address	192.168.1.4
Virtual Sever-1	192.168.2.100
Virtual Sever-2	192.168.2.200
FortiGate VIP-1	100.0.1.100
FortiGate VIP-2	100.0.1.200
External PC1 IP Address	100.0.1.1
External PC2 IP Address	100.0.1.2
External PC3 IP Address	100.0.1.3

Click **Server Load Balance > Application Resources**, and then select the **Persistence** tab.

The screenshot shows the FortiGate web interface. On the left, the 'Server Load Balance' menu is expanded, and 'Application Resources' is selected. On the right, the 'Persistence' tab is active, displaying a table of persistence methods.

Name	Type
LB_PERSIS_SRC_ADDR	Source Address
LB_PERSIS_HASH_SRC_ADDR	Source Address Hash
LB_PERSIS_HASH_SRC_ADDR_PORT	Source Address-Port Hash
LB_PERSIS_HASH_COOKIE	Cookie Hash
LB_PERSIS_SSL_SESS_ID	SSL Session ID
LB_PERSIS_SIP_CALL_ID	SIP Call ID
LB_PERSIS_RDP_COOKIE	RDP Cookie
LB_PERSIS_PASSIVE_COOKIE	Passive Cookie

Click **Create New** to create a new persistence method using the following settings:

Name: **Source-Address-Persistence**

Type: **Source Address**

Timeout (sec): **6**

The screenshot shows the 'Persistence' configuration page in the FortiGate web interface. The 'Name' field is set to 'Source-Address-Persistence', the 'Type' is set to 'Source Address', and the 'Timeout' is set to '6'. The 'Subnet Mask Bits (IPv4)' is set to '32' and 'Subnet Mask Bits (IPv6)' is set to '128'. The 'Match Across Virtual Servers' and 'NAT Source Pool Persistence' options are both disabled. A red arrow points to the 'Save' button.

Persistence	
Name	Source-Address-Persistence
Type	Source Address
Timeout	6
Subnet Mask Bits (IPv4)	32
Subnet Mask Bits (IPv6)	128
Match Across Virtual Servers	☐
NAT Source Pool Persistence	☐

Click **Save**

Click **Server Load Balance > Virtual Server > Virtual Server**. Click the edit icon or double click to edit the new virtual server **WEB-VS**.

Virtual Server				
Content Rewriting Content Routing NAT Source Pool Schedule Pool				
Delete Create New ▼ Search Add Filter				
Name	Type	Address	Port	Profile
WEB-VS	Layer 4	192.168.2.100	80	LB_PROF_TCP
SEC-VS	Layer 7	192.168.2.200	443	LB_PROF_HTTPS

On the **General** tab, change the **Persistence** setting to **Source-Address-Persistence**, and then click **Save**.

Dashboard >

Security Fabric >

FortiView >

System >

Shared Resources >

Network >

Server Load Balance >

Virtual Server

Application Resources

Application Optimization

Real Server Pool

Scripting

SSL-FP Resources

Link Load Balance >

Global Load Balance >

Web Application Firewall >

Network Security >

Virtual Server

Basic **General** Security Monitoring

Configuration

Address

192.168.2.100

Example: 192.0.2.1

Port

80

Default: 80 Range: 0 or 1-65535. You can specify up to eight ports or port ranges, e.g., 80-90 100. Valid values are from 0 to 65535, with 0 for Layer-4 virtual

Connection Limit

0

Default: 0 Range: 0-100000000 concurrent connections

Connection Rate Limit

0

Default: 0 (disabled) Range: 0-86400 connections per second

Interface

port2

Resources

Profile

LB_PROF_TCP

Persistence

Source-Address-Persistence

Method

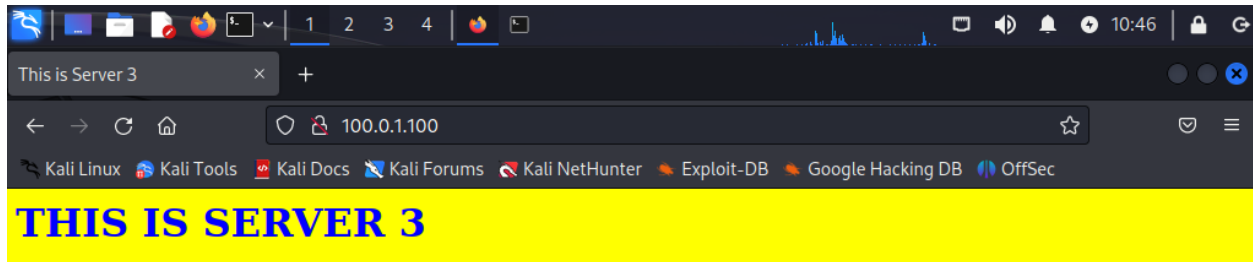
LB_METHOD_DYNAMIC_LOAD

Real Server Pool

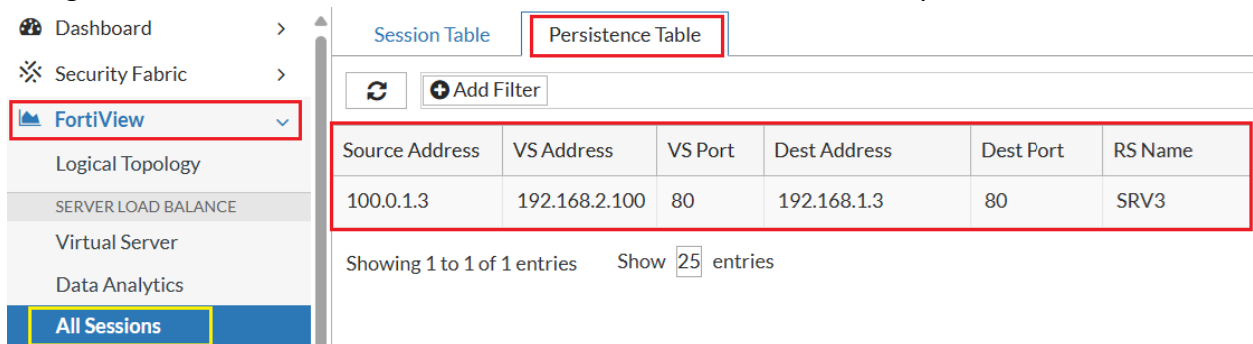
WEB-Pool

Test IP-Based Persistence:

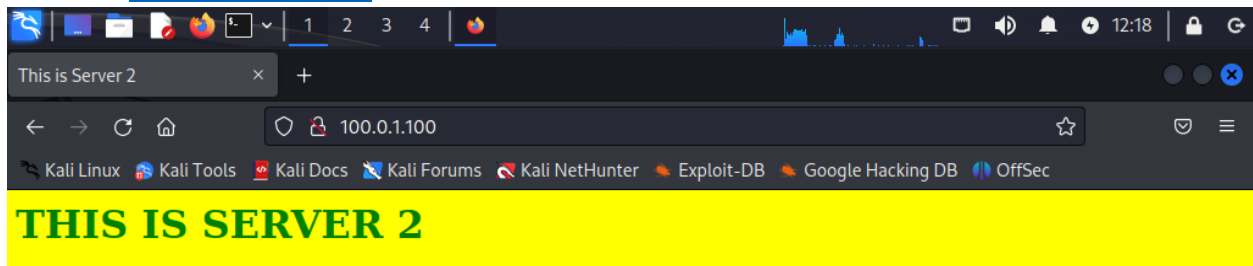
On the External Kali Linux, open Firefox, and then connect to the virtual server at <http://100.0.1.100> . Press **CTRL+SHIFT+R** several times to refresh the browser. Traffic is routed to the same application server.



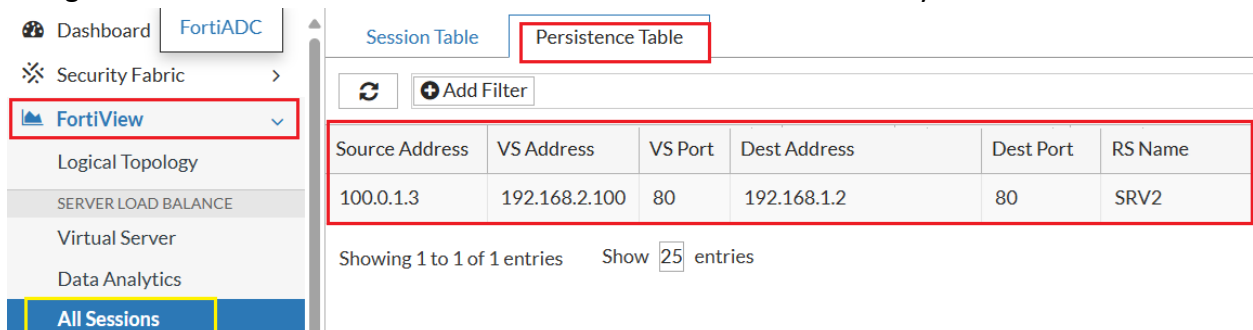
Navigate to **FortiView>All Sessions >** click on **Persistence** Table to verify



Click the browser wait for 6 seconds., open Firefox, and then connect again to the virtual server at <http://100.0.1.100> . Press **CTRL+SHIFT+R** several times to refresh the browser.



Navigate to **FortiView>All Sessions >** click on **Persistence** Table to verify



Navigate to **Log & Report > Traffic Log** can verify the same server in this case **SRV2** is hitting.

Dashboard	>	Download	Refresh	SLB Layer 4&2	25/01/04 12:09:00 - 25/01/06 20:08:40
Security Fabric	>	Add Filter			
FortiView	>				
System	>				
Shared Resources	>				
Network	>				
Server Load Balance	>				
Link Load Balance	>				
Global Load Balance	>				
Web Application Firewall	>				
Network Security	>				
DoS Protection	>				
User Authentication	>				
Log & Report	>				
Traffic Log					

Source	Destination	Virtual Server	Duration (s)	Trans Destination	Real Server
100.0.1.3	192.168.2.100	WEB-VS	7	192.168.1.2	SRV2
100.0.1.3	192.168.2.100	WEB-VS	3	192.168.1.2	SRV2
100.0.1.3	192.168.2.100	WEB-VS	3	192.168.1.2	SRV2
100.0.1.3	192.168.2.100	WEB-VS	3	192.168.1.2	SRV2
100.0.1.3	192.168.2.100	WEB-VS	3	192.168.1.2	SRV2
100.0.1.3	192.168.2.100	WEB-VS	3	192.168.1.2	SRV2
100.0.1.3	192.168.2.100	WEB-VS	3	192.168.1.2	SRV2
100.0.1.3	192.168.2.100	WEB-VS	4	192.168.1.2	SRV2
100.0.1.3	192.168.2.100	WEB-VS	7	192.168.1.3	SRV3

Also can verify by packet capture on FortiADC CLI using command:

diagnose sniffer packet any "port 80: 4

```
FortiADC # diagnose sniffer packet any "port 80" 4
interfaces=[any]
filters=[port 80]
4.416724 port2 in 100.0.1.3.54844 -> 192.168.2.100.80: syn 3449742909
4.416968 port3 out 192.168.1.110.5003 -> 192.168.1.2.80: syn 347321622
4.418393 port3 in 192.168.1.2.80 -> 192.168.1.110.5003: syn 1282607852
4.418409 port2 out 192.168.2.100.80 -> 100.0.1.3.54844: syn 1282607852
4.419429 port2 in 100.0.1.3.54844 -> 192.168.2.100.80: ack 1282607853
4.419440 port3 out 192.168.1.110.5003 -> 192.168.1.2.80: ack 1282607853
4.419626 port2 in 100.0.1.3.54844 -> 192.168.2.100.80: psh 3449742910 a
4.419634 port3 out 192.168.1.110.5003 -> 192.168.1.2.80: psh 347321623
4.420082 port3 in 192.168.1.2.80 -> 192.168.1.110.5003: ack 347322001
4.420093 port2 out 192.168.2.100.80 -> 100.0.1.3.54844: ack 3449743288
4.420481 port3 in 192.168.1.2.80 -> 192.168.1.110.5003: psh 1282607853
4.420490 port2 out 192.168.2.100.80 -> 100.0.1.3.54844: psh 1282607853
4.421301 port2 in 100.0.1.3.54844 -> 192.168.2.100.80: ack 1282608328
4.421311 port3 out 192.168.1.110.5003 -> 192.168.1.2.80: ack 1282608328
4.519896 port2 in 100.0.1.3.54844 -> 192.168.2.100.80: psh 3449743288 a
4.519917 port3 out 192.168.1.110.5003 -> 192.168.1.2.80: psh 347322001
4.520890 port3 in 192.168.1.2.80 -> 192.168.1.110.5003: psh 1282608328
4.520908 port2 out 192.168.2.100.80 -> 100.0.1.3.54844: psh 1282608328
4.522063 port2 in 100.0.1.3.54844 -> 192.168.2.100.80: ack 1282608830
4.522082 port3 out 192.168.1.110.5003 -> 192.168.1.2.80: ack 1282608830
4.550732 port2 in 100.0.1.3.54844 -> 192.168.2.100.80: fin 3449743617 a
4.550753 port3 out 192.168.1.110.5003 -> 192.168.1.2.80: fin 347322330
4.551161 port2 in 100.0.1.3.54856 -> 192.168.2.100.80: syn 710773617
4.551349 port3 out 192.168.1.111.5003 -> 192.168.1.2.80: syn 3940106306
4.551484 port3 in 192.168.1.2.80 -> 192.168.1.110.5003: fin 1282608830
4.551495 port2 out 192.168.2.100.80 -> 100.0.1.3.54844: fin 1282608830
4.551876 port3 in 192.168.1.2.80 -> 192.168.1.111.5003: syn 2226242635
```